

EBPT-CRA: A clustering and routing algorithm based on energy-balanced path tree for wireless sensor networks

Bing Fan^a, Yanan Xin^{b,*}

^a North China Electric Power University, Beijing 102206, China

^b State Grid Xinjiang Electric Power Co., Ltd., Electric Power Research Institute, Urumqi, Xinjiang 830011, China

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ABSTRACT

With the rapid development of internet of things (IoT), the scale of wireless sensor networks (WSNs) is gradually expanding, and it is necessary to study the energy saving and balancing algorithm for WSNs in multi-sensor nodes and large deployment range scenarios. A clustering and routing algorithm based on an energy-balanced path tree (EBPT-CRA) is proposed for large-scale WSNs. First, the energy-balanced path tree (EBPT) is constructed, and then the aggregation betweenness of the nodes are calculated to obtain the cluster head competition coefficients of the nodes. Second, an iterative process is executed to select the cluster heads and build the cluster trees to complete the clustering and intra-cluster routing. Finally, the inter-cluster routing is determined based on the node residual energy and the communication energy. The simulation results demonstrate that the EBPT-CRA can better save and balance the node energy, while improve the network lifetime, throughput and service capability in large-scale WSNs.

1. Introduction

Wireless sensor networks (WSNs) are a very important part of the internet of things (IoT), but wireless sensors are with limited energy, causing the network life cycle bottleneck becomes an important challenge in the development of WSNs (Luo, Zhang & Bai, 2023), especially in IoT scenario where wireless sensors are widely distributed and the environmental conditions do not usually allow for frequent battery replacement or recharging. WSN routing protocols can save network energy effectively, which is divided into planar routing protocols and hierarchical routing protocols. In large-scale WSNs, hierarchical routing protocols are better than planar routing protocols in terms of saving network energy. The basic structure of a WSN under a hierarchical routing protocol is shown as Fig. 1, where the nodes are divided into two categories, cluster heads (CHs) and member nodes. The CHs are responsible for collecting data from the member nodes within their clusters and forwarding the collected data after data fusion. The member nodes are responsible for collecting data from the devices and uploading the collected data to the CHs at a designated time slot. The base station (BS) is responsible for receiving the data forwarded from the CHs and communicating with external networks.

Hierarchical routing algorithms can minimize and balance energy

consumption through the optimization of CH selection, clustering and routing, so many hierarchical routing algorithms have been researched. Some algorithms select CHs and construct clusters based on node energy, node location, and node geometric centrality, while some algorithms perform CH selection and clustering through heuristic algorithms such as genetic algorithm (Alsharafat, Shdaifat, Batiha & Alsukker., 2022), particle swarm algorithm (Wang, Gao, Liu, Sangaiah & Kim, 2019) and improved K-mean algorithm (Wei, Liu, Ding, Deng, Lyu, et al., 2018). The above algorithms can obtain the CHs with high geometric centrality, but there is a phenomenon that the near-BS nodes send data to the far-BS CHs first, and then the CHs transmit data to the BS, increasing the total length of the data transmission path and generating excess energy consumption. CHs consume more energy than member nodes, so relay nodes and multi-hop routing are introduced to balance the energy of CHs. Multi-hop routing avoids long distance communication between nodes, but the near-BS nodes will consume extra energy to forwarding the data coming from far away nodes. Hence the near-BS nodes will die out faster and thus data cannot be sent through them to BS. This area around the BS where communication cannot be further possible is known as “energy hole” (Ranjita and Parul, 2018).

Energy balancing among nodes is conducive to solving the problem of premature death of nodes, increasing the average survival time of

* Corresponding author.

E-mail addresses: fanbing@ncepu.edu.cn (B. Fan), 624042769@qq.com (Y. Xin).

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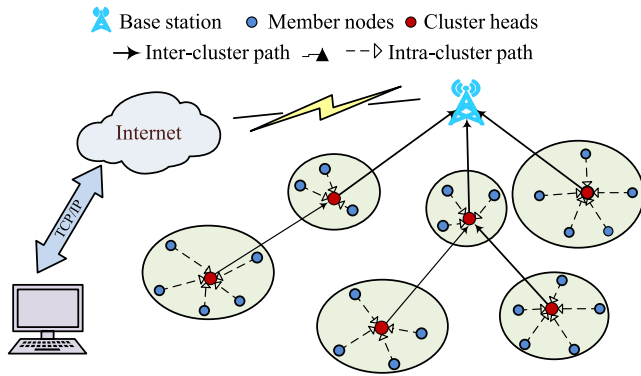


Fig. 1. WSN structure under hierarchical routing protocol.

network nodes, extending network lifetime and improving network throughput, so the research of network energy balancing is of great significance for all energy-constrained WSNs. The research of network energy balancing is also important for wireless recharging sensors network (WRSN) where the nodes are charged via mobile charging devices. However, mobile charging devices can only carry limited energy, and the charging process is complex and subject to a variety of conditions (Zhong, Xu, Zhang, Zhang, & Chen, 2021; Dong, Wang, Li, Cui & Wu, 2019 Lin, Han, Deng & Wu, 2017 Kumar & Mukherjee, 2020; Fanian & Rafsanjani, 2023;). If the energy of nodes in WRSNs is balanced, fewer nodes need to replenish a large amount of energy in a charging cycle, resulting in less constraints and more flexible strategy of recharging, and then making it simpler to keep the network functional.

In large-scale WSNs, due to the large number of nodes and their wide distribution, the near-BS CHs and the near-BS forwarding nodes take on a large amount of data forwarding work, which accelerates the energy of the near-BS nodes and exacerbates the “energy hole” problem. Therefore, the study of network energy balancing is even more important and challenging in large-scale WSNs.

To address the above problems, we have proposed a clustering and routing algorithm FC-CRA recently (Fan & Xin, 2023), which can alleviate the “energy hole” problem effectively, but we find that the network energy is mainly consumed in data transmission in large-scale WSNs, and the idea of clustering firstly and routing secondly cannot maximize energy utilization due to the constraints on routing imposed by the clustered network structure. However, if the routes that have a great impact on the node energy are optimized first, and then clustering is performed based on the routes, the node energy will be better saved and balanced. Therefore, this paper introduces the idea of routing firstly and clustering secondly into a new clustering and routing algorithm based on energy-balanced path tree (EBPT-CRA) for large-scale WSNs. The main contributions of this paper are as follows:

(1) An energy-balanced path tree (EBPT) is constructed according to the residual energy and data transmission energy of each node. Based on the EBPT, a concept of node convergence betweenness (NCB) is proposed to calculate the cluster head competition coefficient (CHCC), through which the CHs are selected one by one. This method selects the CHs with a low geometric centrality, which can avoid backward data transmission and reduce the path length of data transmission.

(2) A clustering algorithm based on the CHCC and cluster head attraction (CHA) is proposed to save and balance network energy. This algorithm clusters WSN nodes in batches and completes intra-cluster routing, in which, some nodes are clustered based on the selected CHs and the current EBPT, while some nodes are clustered based on the CHA.

(3) For inter-cluster communication, the near-BS nodes are selected as forwarding nodes to transmit the data from the CHs. This method provides multiple paths for data transmission between the CHs and BS, which decentralizes the forwarding task of the CHs near the BS. When the residual energy of a CH or forwarding node drops to $x\%$ of its

original residual energy, the WSN updates clusters and routing.

(4) In this paper, we propose the idea of routing firstly, clustering secondly and adjusting the route thirdly to achieve the clustering and routing of WSNs, instead of the method of clustering firstly and then routing, avoiding the problem of mutual interference between clustering and routing, achieving the optimization of the overall clustering and routing, improving node load balance and alleviating the “energy hole” problem.

This paper is organized as follows: Section 2 introduces the related works, Section 3 describes the system model, Section 4 provides a detailed description of the proposed algorithm EBPT-CRA, Section 5 simulates and analyses the EBPT-CRA to verify the feasibility and superiority of the EBPT-CRA, and Section 6 gives the conclusions of this paper.

2. Related works

2.1. CH selection and communication strategy

The classical LEACH protocol Jesudurai & Senthilkumar, 2019; Devi, Priya & Priya, 2020) has strong randomness in CH selection, is limited by single-hop communication strategy in network scalability, and does not achieve better results in network energy balance, so many scholars have conducted research on the improvement of the LEACH protocol.

In order to enhance the reasonability of CH selection, (Fang et al., 2021) proposed an improved LEACH protocol named LEACH-TM, which selects CHs based on the number of dead nodes. The number of CH nodes, residual energy and density of neighbor nodes are considered to constrain the size of clusters and improve energy efficiency. Meanwhile, the trust management scheme is introduced into the LEACH-TM to defend against internal attacks. To improve energy efficient, a TEECH algorithm (Kumar, Rani, Kumar, Verma & Koundal, 2023) is proposed to improve the balance of cluster distribution. However, the number of clusters is fixed, which can't quickly adapt to the change in network structure caused by the dead nodes. Considering static sink node (SSN) and mobile sink node (MSN), (Amutha, Sharma and Sharma, 2022) proposed the HBAC algorithm for CH selection and routing, in which five parameters, i.e., node residual energy, intra-cluster distance, CH-to-SSN/MSN distance, degree of node and node centrality are considered to save energy and prolong network lifetime. However, due to the limitation of individual code length, the number of CHs is fixed, and single-hop communication is used with clusters, which cannot adapt to the change of the distribution of surviving nodes in large-scale WSNs. (Baradaran & Navi, 2020) proposed a HQCA-WSN algorithm which uses a fuzzy algorithm considering node energy and node-to-BS distance for selecting CHs, and accordingly presents a cluster quality index based on the cluster condition and the impact of single node entry to obtain high quality clusters. However, the computation complexity is large when there are more network nodes, and the rationality of simply using node energy and node-to-BS distance as CH selection criteria is not high. A CH selection algorithm using the fuzzy C-means is presented to estimate the number of CHs based on the number of nodes in the network (Su & Zhao, 2018). The node-to-CH distance and the number of nodes in the cluster are as the basis for node entry, excluding the influence of extreme nodes on the clustering results. However, the number of CHs in this algorithm is estimated under a uniform distribution of nodes, which is actually non-uniform, so the results are not very accurate. (Gbadouissa, Ari, Titouna, Gueroui, & Thiare, 2020) combined hypergraph theory and a heuristic algorithm to obtain the optimal clustering scheme, and selected the node with the largest value of the CH selection function within the hyper-edge as the CH, but the algorithm did not consider the influence of node location on node communication, the local optimum trap, and the computation complexity.

The energy balance of nodes is not only affected by the clustering results, but also by the communication routing strategy. In a large-scale WSN, single-hop communication strategy causes long-distance

communication, which accelerates node energy consumption. To solve this problem, the multi-hop communication strategy was introduced into WSNs. (Darabkh, Odetallah, Al-qudah, Khalifeh, & Shurman, 2019) divided the network into layers, then divided the network into sub-layers according to the distance from the center of each layer to the BS, and built clusters based on the size of the sub-layers and a cluster merging algorithm to avoid very small clusters. At the same time, they reduced long distance transmission by specifying that only adjacent layers communicate with each other, but when the span of each layer is small, the number of data forwarding increases, generating excess energy consumption, and in addition, cluster merging can generate very large clusters, resulting in uneven energy distribution between clusters. A COCA algorithm Li, Liu, Chen, Jia, Li and Xiong (2013) is proposed, which divides the network into layers and sub-layers to construct clusters. The algorithm selects the next hop according to the remaining energy of the nodes, and reduces the cluster size by increasing the number of clusters in the near-BS area to spread the data forwarding pressure of the nodes, but generates excess energy by forwarding frequently in the dense area of CHs. (Randhawa and Jain, 2019) proposed an MLBC algorithm, which introduces a fuzzy algorithm and a multi-objective function based on path reliability and energy efficiency to find the optimal routing, but the performance indexes and the applicability of the objective function differ in different scenarios. (Vinitha, Rukmini & Shirajsumehra, 2019) accomplished the optimization of inter-cluster communication paths by means of the Taylor C-SSA algorithm and used multiple indexes to measure the path quality, but the more indexes considered by this algorithm the higher the computational complexity. (Bhardwaj and Kumar, 2019) proposed a MOFPL protocol, which is an energy-aware routing protocol based on a multi-objective function that considers energy, distance, node data generation rate, time delay and cluster density. However, the protocol uses heuristic algorithms in both the clustering and routing phases, resulting in a long execution time and high complexity of the protocol. For intra-cluster communication, (Wang, Shen, Vijayakumar, Zhu & Tian, 2019) add random numbers to the data transmission process to improve communication security, and reduces the amount of similar data transmitted through data identification. (Amuthan and Arulmurugan, 2021) proposed an algorithm combining a prediction algorithm with semi-Markov chain to establish inter-cluster communication paths, which takes into account security and reliability factors, but lacks consideration for energy efficiency. Aiming at “energy hole” problem, (Yang, 2022) proposed an elliptic guided routing algorithm, which combines the network conditions and landmark to determine the elliptic guided communication paths between nodes to ensure data transmission when “energy holes” occur in the network. (Fan & Xin, 2023) proposed a clustering and routing algorithm FC-CRA, which constructs clusters based on adaptive cluster radius for the fast changes of large-scale WSNs in IoT, introduces path energy function into the intra-cluster routing, and uses some near-BS nodes as forwarding nodes in the inter-cluster routing, improving node load balance and alleviating the “energy hole” problem.

2.2. Energy balancing algorithms

In the multi-hop communication mode, the near-BS nodes are subjected to more forwarding tasks, resulting in an uneven distribution of energy consumption. To achieve a balanced energy consumption, a dual CH model is introduced, where one CH is responsible for data fusion and the other CH is responsible for data forwarding (Jesudurai and Senthil Kumar, 2019; Panag & Dhillon, 2018). From another point of view, non-uniform clustering can also lead to energy balance for that the near-BS cluster which takes on more data forwarding should have less member nodes, and conversely, the far-BS cluster which takes on less data forwarding can have more member nodes. Arjunan *et al.* (Arjunan & Sujatha, 2019) classified some of the non-uniform clustering algorithms into probabilistic, deterministic and preset categories. Non-uniform clustering can limit the size of clusters through cluster radius,

network region hierarchy, and so on. A non-uniform cluster radius and sleep-wake cycle mechanism is proposed (Liu, Wu, Chen & Gui, 2010) to achieve energy saving, but additional energy is required to start and wake up the sleeping nodes, and the switching time and switching frequency of the node state need further consideration. A fuzzy algorithm is introduced to evaluate the number of CHs and achieve load balancing among CHs by adjusting the cluster radius, but the calculation of cluster radius in the algorithm takes fewer factors into account (Bagci & Yazici, 2013; Zhang, Li, Yan & Li, 2011) proposed an LCA-DG algorithm, which limits the cluster size and achieves energy balancing through an optimization threshold and a reception threshold, but the algorithm does not consider the energy consumption of data forwarding. (Din, Rehman, Ullah, Park, Kim & Kim, 2022) divide each cluster into several sub-clusters, and uses the Analytical Hierarchy Process (AHP) to select sub-cluster heads (SCH), reducing redundant data transmission, reducing the load of CHs, and improving the load balancing of nodes in the network. In order to balance the energy consumption between CHs and member nodes, (Lu *et al.*, 2022) proposed an ASHEED algorithm, in which, when the energy of CH is less than the energy threshold, a member node is selected as an agent based on the energy to perform data fusion and transmission, balancing the energy consumption between CH and its member nodes. The agents rotate within the cluster until no node meets the energy threshold, and then the CH will perform data fusion and transmission again. However, how to combine routing strategies to save and balance energy is not discussed.

In terms of energy saving, fixed-period cluster reconfiguration generates more control information and a larger amount of energy consumption, so the frequency of cluster reconfiguration should be reduced to save energy. (Wang *et al.*, 2019) update clusters based on the CH validity and node remaining energy. (Radhika and Rangarajan, 2019) use a fuzzy system to obtain the update period based on the data rate of nodes, and the cluster reconfiguration is performed based on the update period. In (Panchal & Singh, 2021), re-clustering is performed in a few rounds to avoid nodes sending unnecessary control packets to the BS. In (Guleria, Verma, Goyal, Sharma Benslimane & Singh, 2021), clusters are updated when any wireless link is fault, which ensures link connectivity as well as saves energy.

2.3. Applications of path tree in WSNs

Path trees have been introduced into WSNs, which can obtain the optimal solutions for WSN communication paths considering different factors by defining different edge weights. A shortest path tree is constructed based on the Prim algorithm and Dijkstra algorithm to reduce the number of nodes in data transmission, save network energy and reduce network delay. In the parent node reselection process, the authors considered the number of data retransmissions, communication quality and node residual energy, but did not consider the parent node load, resulting in a less balanced network energy (Yao, Wang, Sun & Qiu, 2007). To extend the network lifetime and increase the network throughput, (Karunanithy and Velusamy, 2020) proposed a cluster-tree based energy efficient data gathering (CTEEDG) protocol, which establishes a tree topology between clusters with BS as the root node during the inter-cluster communication phase to guarantee the availability of the congestion-free shortest path from each cluster to the BS. (Delaney, Higgs & Hare, 2014) proposed a NH (neighbourhood heuristic) based parent node selection method to reduce the impact of parent node failure on the network links, and verified the path stability under this method by two performance indexes, route persistence and route universality. (Alinia, Hajiesmaili & Khosari, 2015) constructed a dynamic fusion tree using Markov model under transmission delay constraints to adjust the parent nodes, reduce the transmission delay and improve the data transmission quality. To minimize the broadcast path length, (Gong, Fu, Fu, Zhao Wang & Wang, 2016) proposed a toward source tree (TST), proved that the TST length is at the same level as Steiner tree, and verified that the structure of the TST is the best broadcast tree structure.

The effectiveness of the TST algorithm for improving the network energy efficiency is verified by two indexes, information complexity and the amount of forwarded data. (Yigit, Incel & Gungor, 2014) proposed a receiver-based channel selection method and evaluated the WSN performance based on different trees which include routing trees, capacitated minimum spanning trees (CMSTs), and minimum hop spanning trees (MHSTs). To find the inter-cluster communication path, (Yu et al., 2019) build a forwarding tree based on the rule that the node with the shortest distance and largest residual energy is selected as the next hop of current node.

The above works use trees to solve the energy consumption problems, data transmission reliability or real-time problems, and so on, but does not consider using trees to solve the clustering problem.

2.4. The state of the art

The state-of-the-art WSN clustering and routing technologies are listed in Table 1. The main shortcoming is that the cluster structure and routing strategy cannot adapt to the changes in energy distribution or

cannot save energy better in large-scale or sparse WSNs in the IoT, which leads to the load unbalance and the “energy hole” problem. So that, long-term and efficient data transmission is difficult to guarantee. On the other hand, since network energy is mainly consumed in data transmission in large-scale WSNs, the idea of clustering firstly and routing secondly cannot maximize energy utilization due to the mutual interference between clustering and routing. The proposed EBPT-CRA introduces the idea of routing firstly and clustering secondly to overcome these problems. In this algorithm, the EBPT is firstly constructed based on node energy consumption and residual energy, and then CHs are selected considering node load, distance, and energy, which balances the intra-cluster energy consumption. The intra-cluster multi-hop routing strategy is more suitable for large-scale WSNs. The inter-cluster routing strategy based on energy and load distribution can balance the inter-cluster energy consumption better.

Table 1
The state-of-the-art clustering and routing technologies for WSNs.

	Clustering	Routing	Cluster updating strategy	Limitations
EC-PSO (Wang et al., 2019)	The PSO algorithm is used to select CHs, and number nodes join the nearest CH.	The optimal single-hop distance and the ant colony are applied on the intra-cluster and inter-cluster communication respectively.	After 50 rounds, the cluster is updated if a dead node appears, a CH is invalid, or the number of rounds reaches 20 each time.	When there is no dead node or invalid CH, CH-updating is with fixed rounds, which is not suitable for rapid adjustment of energy distribution. The intra-cluster communication method is prone to isolated nodes in large-scale WSNs.
TEEECH (Kumar, Rani, Kumar, Verma, & Koundal, 2023)	The CHs are selected on different layers based on an improved LEACH, and the clusters are built by fuzzy C-mean algorithm.	The CHs communicate with the BS directly.	Clusters are updated each round.	The single hop communication strategy consumes much energy in data transmitting, which is not suit for large-scale WSN.
HQCA-WSN (Baradaran and Navi, 2020)	A fuzzy algorithm considering node energy and node-to-BS distance is applied to select CHs, and a cluster quality index is introduced to decide whether a single node should join the cluster or not.	Single hop communication is adopted in data transmission phase.	Clusters are updated each round.	The single hop communication strategy is not suit for large-scale WSN, and the energy balance in data routing is not considered.
HGC (Gbadouissa et al., 2020)	The nodes are evenly divided into clusters according to the given number of CHs based on the hypergraph theory and a heuristic algorithm.	Member nodes communicate with the CH directly, and the inter-cluster routing is based on the minimum communication energy tree.	Clusters are updated every set time τ .	The cluster coverage will increase as the number of surviving nodes decreases, leading to a higher energy consumption, and the inter-cluster routing is lack of consideration about load balance.
ASHEED (Lu et al., 2022)	The temporary CHs are elected based on hybrid energy-efficient distribution, and then the final CHs and clusters are determined by the competition radius. Meanwhile, rotating agent nodes are introduced to balance the energy consumption.	Single hop communication is adopted for inter-cluster and intra-cluster communication.	When a new dead node appears, the clusters are updated.	Long-distance communication consumes a lot of energy, so inter-cluster communication with single hop is not suit for large-scale WSNs.
CTEEDG (Karunanithy and Velusamy, 2020)	The tentative CHs are selected by the LEACH algorithm, and the final CHs are determined by a competition mechanism based on broadcasting time which considers different metrics such as residual energy, number of neighbours, and average distance.	Routing based on the tree with the BS as the root node to guarantee the congestion-free shortest path from each CH to the BS.	Clusters and routing are updated each round.	High update frequency leads to unnecessary energy overhead.
UCRA-GSO (Yu et al., 2019)	The CHs and clusters are determined by the glowworm swarm optimization (GSO) algorithm which considers the CH local density, the average distance and energy consumption within a cluster, and the dispersibility of CHs.	The node with the shortest distance and largest residual energy is selected as the next hop of current node.	Clusters and routing are updated each round.	High update frequency leads to unnecessary energy overhead, and only consider the next hop node leads to the node load unbalanced.
FC-CRA (Fan and Xin, 2023)	The clusters are constructed by an iterative process based on the adaptive cluster radius that considers node energy and location.	The intra-cluster routing maximizes the minimum node residual energy, and some near-BS nodes are introduced into the inter-cluster routing to alleviate the “energy hole” problem.	When the residual energy of a CH drops to 50 % of its original residual energy, the WSN starts the clustering process again.	As other algorithms, the idea is also clustering firstly and routing secondly, which cannot avoid the mutual interference between clustering and routing, and utilize network energy better.

3. System model

3.1. Network model

In this paper, the WSN consists of a large number of randomly distributed nodes and a BS locates at the center. The network model is shown as Fig. 2.

The CHs are responsible for collecting the data from their member nodes and data fusion to reduce the amount of forwarded data. The relay nodes are responsible for forwarding the data from the CHs and other relay nodes. All sensor nodes are homogeneous.

3.2. Energy consumption model

The energy consumption model for WSN in this paper is the first-order radio model (Wang et al., 2019), which is shown in Fig. 3.

The formula of transmitting energy consumption E_{TX} is

$$E_{TX}(l, d) = \begin{cases} l(E_{elec} + \varepsilon_{fs}d^2), & d < d_0 \\ l(E_{elec} + \varepsilon_{mp}d^4), & d \geq d_0 \end{cases} \quad (1)$$

where, l is the amount of data, d is the distance between the transmitter and receiver, E_{elec} is the energy consumption for processing one bit data in transmitter or receiver, ε_{fs} and ε_{mp} are attenuation coefficients for free space channel and multi-path propagation channel, d_0 is the distance threshold for the energy consumption model and can be calculated as follows:

$$d_0 = \sqrt{\frac{\varepsilon_{fs}}{\varepsilon_{mp}}} \quad (2)$$

The formula of receiving energy consumption E_{RX} is

$$E_{RX}(l) = lE_{elec} \quad (3)$$

4. Clustering and routing algorithm based on EBPT

Considering CH selection and data forwarding paths, this paper proposes a clustering and routing algorithm, which first constructs an EBPT, then constructs and updates clustering trees (CTs) based on the EBPT to obtain the clusters and the intra-cluster communication paths, and finally plans inter-cluster communication paths based on the node loads. The symbols of the parameters involved in the EBPT-CRA are summarized in Table 2.

4.1. EBPT construction

To achieve energy balance among network nodes, this paper uses the shortest path algorithm that considers the energy of data transmission and the residual energy of nodes to build the EBPT. First, all nodes in the network are sorted in descending order according to the node-to-BS distance to obtain node sequence Q . When there are multiple nodes with equal node-to-BS distance, these nodes are arranged in random order. Second, the edge weight w_{ij}^t between node i and node j in the t -th iteration is calculated, which is defined as

$$w_{ij}^t = \frac{E_{TX}(l, d_{ij})}{E^t(i)} + \frac{E_{RX}(l)}{E^t(j)} \quad (4)$$

where $E_{TX}(l, d_{ij})$ is the energy consumption of node i to send l bit data to node j , $E_{RX}(l)$ is the energy consumption for node j to receive l bit data, $E^t(i)$ and $E^t(j)$ are the residual energy of node i and node j in the t -th iteration respectively. If node j is the BS, $E^t(j) = \infty$. Then, the iterative process of constructing the EBPT begins.

In Step 1, taking the first node Q_1 in Q as the source node and the BS as the destination node, the shortest path algorithm Dijkstra is used to calculate the minimum weight path S_t for node Q_1 to communicate with the BS, and add S_t to the EBPT. In Step 2, all nodes in S_t are removed from Q , and the residual energy of all nodes in S_t is updated assuming that node Q_1 transmits data along S_t . In Step 3, if the EBPT contains all nodes in the network, the iterative process ends and the EBPT is output, otherwise Setp1 is returned. The detailed description of the process is shown in Process 1.

Process 1 EBPT construction

Input: node location, energy and data information

Output: EBPT

```

1:  $t \leftarrow 1, EBPT \leftarrow \emptyset, T_n \leftarrow \phi;$ 
2: FOR all nodes
3:   obtain node set  $Q$  by sorting nodes in descending order according to the
   node-to-BS distance, and calculate  $w_{ij}^t$ ;
4: END FOR
5: WHILE  $T_n \neq V$ 
6:   calculate the shortest path  $S_t$  from node  $Q_1$  to BS by Dijkstra algorithm;
7:    $T_n \leftarrow T_n \cup$  all nodes in  $S_t$ ;  $EBPT \leftarrow EBPT \cup S_t$ ;
8:   remove all nodes in  $S_t$  from  $Q$ ;
9:   FOR all node  $i$  in  $S_t$ 
10:    update residual energy  $E^t(i)$  after node  $Q_1$  transmitting data along  $S_t$ ;
11:   END FOR
12:    $t \leftarrow t + 1$ ;
13: END WHILE
```

Taking a 20-node network as an example, the EBPT is constructed as

→ Inter-cluster communication path ---→ Intra-cluster communication path
 Base station  Member nodes  Cluster heads  Relay nodes

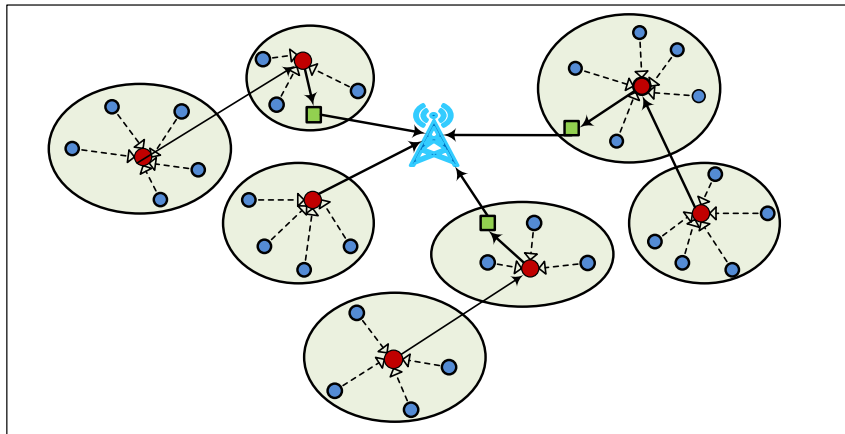


Fig. 2. Network model.

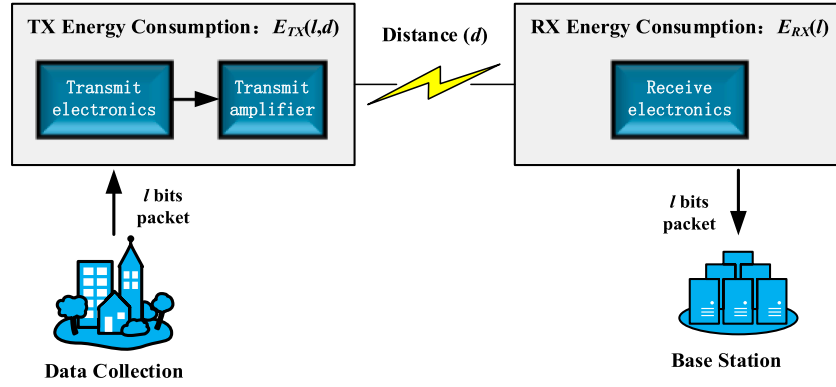


Fig. 3. WSN energy consumption model.

Table 2

Description of the symbol definition.

Symbol	Definition
t	Number of iterations
V	Network node set
N	Number of network nodes
T_n	Node set in EBPT
Q	Node ID sequence
P	Percentage of CHs
CT	Set of clustering trees
HS	Set of cluster heads
W_{ij}^t	Weight of edge $e(i,j)$ in the t -th iteration
d_{ij}	Distance between node i and node j
E_0	Node initial energy
$E(i)$	Residual energy of node i

shown in Fig. 4.

4.2. CT construction

In order to improve the energy balance of the network, this section selects the CHs and constructs the CTs based on the EBPT to complete the cluster construction and intra-cluster routing.

4.2.1. First-stage CT construction

The load distribution is a key factor affecting the energy balance of the network. To quantify the importance of nodes base on the load distribution in the EBPT, the node convergence betweenness (NCB) of

node i is defined as

$$C(i) = \frac{m(i)}{N} \quad (5)$$

where $m(i)$ is the number of all children nodes in the current EBPT with node i as the root node. Based on the NCB, the first-stage cluster head competition coefficient (CHCC) of node i is defined as

$$P_C^1(i) = \begin{cases} e^{-\frac{1}{2}(\frac{C(i)}{C_0} - 1)^2}, & |C(i) - C_0| \leq \delta C_0 \\ 0, & |C(i) - C_0| > \delta C_0 \end{cases} \quad (6)$$

where

$$C_0 = \frac{1}{NP} \quad (7)$$

is the NCB of the CHs when the network is uniformly clustered according to the CH percentage P . In order to avoid that the nodes with too large or too small NCB become CHs resulting in very large and very small clusters, a constraint $|C(i) - C_0| \leq \delta C_0$ is used to limit the CH selection range, and in this paper $\delta = 1/3$.

The construction of the first-stage CTs is an iterative process. First, the first-stage CHCC of each node in the current EBPT is calculated. If there exists some first-stage CHCCs greater than 0, the node with the maximal first-stage CHCC is selected as the CH in this iteration, and the subtree in the current EBPT with this CH as the root node is a CT. Then, the CT is removed from the current EBPT. The above iterative process is repeated until the number of the CHs reaches NP , all nodes in the

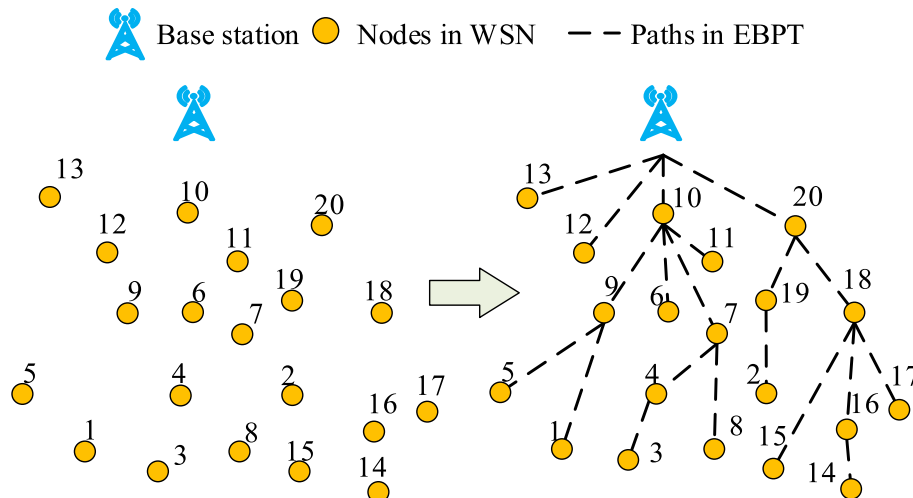


Fig. 4. An EBPT presentation.

network are clustered, or there are no first-stage CHCCs greater than 0. The detailed description of the construction process is shown in **Process 2**.

Each CT is a cluster, where the root node is the CH, and the paths in the tree are intra-cluster communication paths. Since the near-BS nodes are mostly with a large NCB, they hardly become CHs during the construction of the first-stage CTs under the constraint, which alleviates the “energy hole” problem.

Based on the EBPT shown in Fig. 4, when the percentage of CHs $P=0.2$, i.e., four CHs need to be selected, the first-stage CHCC of each node is calculated as shown in Table 3. The process of constructing the first-stage CTs and updating the EBPT based on **Process 2** are shown in Fig. 5.

Process 2 First-stage CT construction

Input: EBPT, N , P

Output: first-stage CT

```

1:      CT ← φ, HS ← φ, Flag ← 1;
2:      WHILE Flag = 1
3:          Pmax ← 0;
4:          FOR all node  $i$  in EBPT
5:              IF  $P_C^1(i) > P_{\max}$ 
6:                  Pmax ←  $P_C^1(i)$ ,  $k \leftarrow i$ ;
7:              END IF
8:          END FOR
9:          IF Pmax = 0
10:             Flag ← 0;
11:          ELSE
12:             obtain the subtree CTk with node  $k$  as the root;
13:             HS ← HS ∪ { $k$ }; CT ← CT ∪ {CTk};
14:             EBPT ← EBPT − {CTk};
15:             IF EBPT = {BS} or |HS| =  $N \cdot P$ 
16:                 Flag ← 0;
17:             END IF
18:          END WHILE

```

4.2.2. Second-stage CT construction

After the first-stage CTs are constructed, if there are un-clustered nodes in the network and $|HS| < N \cdot P$, this paper constructs the second-stage CTs with appropriately relaxed constraints based on the distribution, energy and load of un-clustered nodes. In the EBPT, the near-BS nodes are easy to become heavily loaded nodes because they need to forward more data. Therefore, the construction process of the second-stage CTs should try to select the nodes with high residual energy, far from the BS, and with the NCB satisfying the constraint as the CHs. Define the second-stage CHCC of un-clustered node i as

$$P_C^2(i) = \begin{cases} \left(\frac{E(i)}{E_0} + \frac{d_{iBS}}{\sqrt{x^2 + y^2}} \right) e^{-\frac{1}{2} \left(\frac{C(i)}{C_0} - 1 \right)^2}, & |C(i) - C_0| \leq \theta C_0 \\ 0, & |C(i) - C_0| > \theta C_0 \end{cases} \quad (8)$$

where x and y are the side lengths of the rectangular monitoring region respectively, and $|C(i) - C_0| \leq \theta C_0$ is the constraint for the second-stage CH selection. Let $\theta > \delta$ to relax the constraint, and in this paper $\theta = 1/2$. The process of the second-stage CT construction is similar to

Table 3

NCB and first-stage CHCC of each node in Fig. 4.

Node ID	$C(i)$	$P_C^1(i)$	Node ID	$C(i)$	$P_C^1(i)$
1	0.05	0	11	0.05	0
2	0.05	0	12	0.05	0
3	0.05	0	13	0.05	0
4	0.10	0	14	0.05	0
5	0.05	0	15	0.05	0
6	0.05	0	16	0.10	0
7	0.20	0.980	17	0.05	0
8	0.05	0	18	0.25	1.00
9	0.15	0	19	0.10	0
10	0.50	0	20	0.40	0

Process 2, except that the input is the set EBPT, CT and HS after the execution of **Process 2**, and $P_C^1(i)$ is replaced by $P_C^2(i)$, so the other parts of the process are not repeated here. After the second-stage CT construction, the clustering results of the network in Fig. 4 and the updated EBPT are shown in Fig. 6.

4.2.3. Final-CT construction

When the second-stage CT construction is completed, if there are still un-clustered nodes in the network, these nodes are clustered based on the CHA of the CHs. When a node chooses a cluster to join, it should consider the energy of the CH, the node-to-CH distance, and the CH load. Define the CHA of CH k to node i as

$$P_{CA}(k, i) = \begin{cases} \left(\frac{E(k)}{E_0} + \frac{d_0}{d_{ki}} \right) P_C^1(k, i), & d_{iBS} > d_{ik} \\ 0, & d_{iBS} \leq d_{ik} \end{cases} \quad (9)$$

where

$$P_C^1(k, i) = e^{-\frac{1}{2} \left(\frac{C(k) + C(i)}{C_0} - 1 \right)^2} \quad (10)$$

Based on the CHA, the process of un-clustered nodes joining into clusters is shown in **Process 3**. After the final CTs are formed, the paths in the final CTs are the intra-cluster communication paths, and the remained un-clustered nodes communicate with BS along the paths in the initial EBPT obtained by **Process 1**.

Process 3 Final-CT Construction

Input: Current EBPT and CT

Output: Final CT

```

1:      WHILE EBPT ≠ {BS}
2:          FOR every node  $i$  in EBPT
3:              Pmax ← 0,  $I \leftarrow 0$ ,  $K \leftarrow 0$ ;
4:              FOR every CH  $k$ 
5:                  calculate  $PCM(k, i)$ 
6:                  IF  $P_{CA}(k, i) > P_{\max}$ 
7:                      Pmax ←  $PCM(k, i)$ ,  $I \leftarrow i$ ,  $K \leftarrow k$ ;
8:                  END IF
9:              END FOR
10:             IF Pmax ≠ 0
11:                 CTK ← CTK ∪ { $I$ }, EBPT ← EBPT − { $I$ };
12:                 the node closest to node  $I$  in CTK is the next hop of node  $I$ ;
13:             ELSE
14:                 all node  $i$  in EBPT communicate with the BS along the paths in the
15:                 initial EBPT obtained by Process 1;
16:                 EBPT ← {BS};
17:             END IF
18:         END WHILE

```

4.3. Inter-cluster routing

In the inter-cluster communication, the nodes that are non-CH nodes and can reach the BS with a single-hop in the initial EBPT are selected as forwarding nodes to share the data forwarding pressure for the near-BS CHs and alleviate the “energy hole” problem.

To perform inter-cluster routing, a weight matrix is first constructed based on the energy, load and location information of the CHs and forwarding nodes, and then inter-cluster routing is calculated by the shortest path algorithm Dijkstra. In the weight matrix, W_{ij} denotes the weight between nodes i and j involved in the inter-cluster communication, which defined as

$$W_{ij} = a \frac{E_{TX}(L_i, d_{ij})}{E(i)} + b \frac{E_{RX}(L_i)}{E(j)} + c \frac{E_{TX}(L_j, d_{jBS})}{E(j)} \quad (11)$$

where L_i is the load of node i , which is the amount of data generated by node i itself, L_j is the load of node j , including the amount of data generated by node j itself and the amount of data from node i . If node j is the BS, $L_j = 0$ and $E(j) = \infty$, a , b and c are constant coefficients satisfying

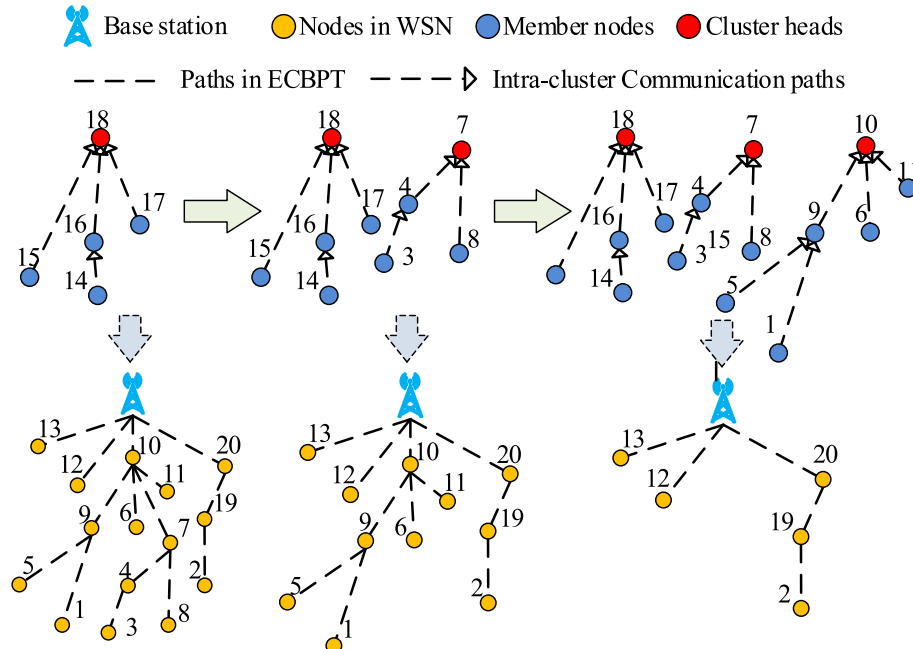


Fig. 5. Demonstration of Process 2.

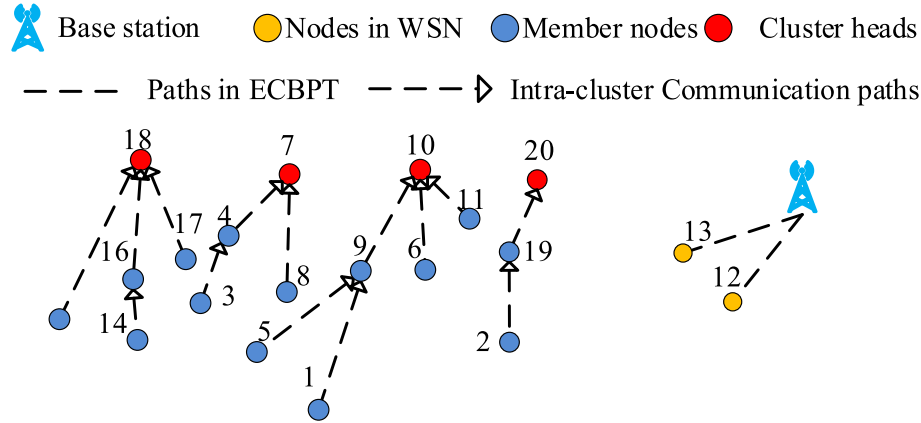


Fig. 6. CTs and EBPT after the second-stage CT construction.

$a + b + c = 1$. The first term of Eq. (11) reflects the residual energy of node i and the energy consumption of transmitting its load, which is the main factor of W_{ij} , so in this paper, $a = 0.6$. The second term of Eq. (11) reflects the residual energy of node j and the energy consumption of receiving the data from node i , which is independent of the distance between node i and node j , and has less influence on W_{ij} , so in this paper, $b = 0.1$. The third term of Eq. (11) is to reduce the transmission of data in the direction away from the BS at node i , whose influence on W_{ij} cannot be neglected, so in this paper, $c = 0.3$.

5. Simulation and analysis

The proposed EBPT-CRA firstly constructs the initial EBPT according to **Process 1**, secondly obtains the CTs through the tree stages in terms of **Process 2** and **Process 3**, and finally completes the intra-cluster routing and inter-cluster routing. In order to verify the superiority of the EBPT-CRA, this paper compares the EBPT-CRA with the LEACH, EC-PSO (Wang et al., 2019), HGC (Gbadouissa et al., 2020), TEECH (Kumar et al., 2023) and our previous algorithm FC-CRA (Fan et al., 2023) on the platform of MATLAB R2019a for simulation and analysis,

and the comparative performance indicators are as follows:

- Number of dead nodes: the number of nodes with zero residual energy.
- Network residual energy: the sum of the remaining energy of the surviving nodes.
- Accumulative network throughput (ANT): the sum of the cumulative number of packets received by the BS.
- Average node lifetime (ANL): the average number of surviving rounds for all nodes.
- Network lifetime (NL): the number of network operation rounds when 50 % of the network nodes are dead.
- Network service capability (NSC): The total number of packets or the amount of data collected by the BS during the NL.

When the residual energy of a CH or forwarding node drops to 50 % of its original residual energy, the WSN updates the clusters and routing. The average of 50 simulation results was taken for comparison and analysis, and the other simulation parameters are shown in Table 4. The comparison results are shown in Fig. 7 to 10.

Table 4
Simulation parameters.

Parameter	Value
Length of the simulation square area	300 m
The number of uniformly distribution sensor nodes	100
Data compression ratio	0.4
Initial energy of nodes in WSN E_0	1.0 J
Energy consumption for processing one bit data in transmitter or receiver E_{elec}	50 nJ/bit
Attenuation coefficient of free space channel ϵ_{fs}	10 pJ/(bit·m ²)
Attenuation coefficient of multi-path propagation channel ϵ_{mp}	0.0013
Energy consumption for compressing one-bit data at a CH E_{da}	5 nJ/bit
Compression ratio c	0.4
Data packet length	2500 bit
Control packet length	20 bit
Data generation rate	2500 bit/round
CH percentage P	10 %

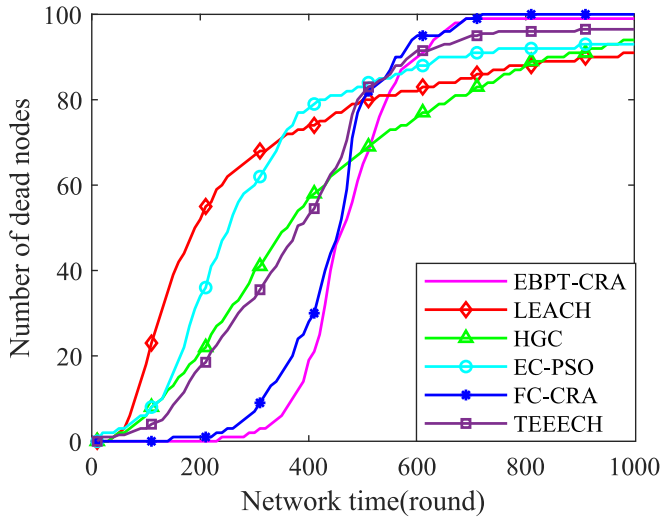


Fig. 7. Number of dead nodes.

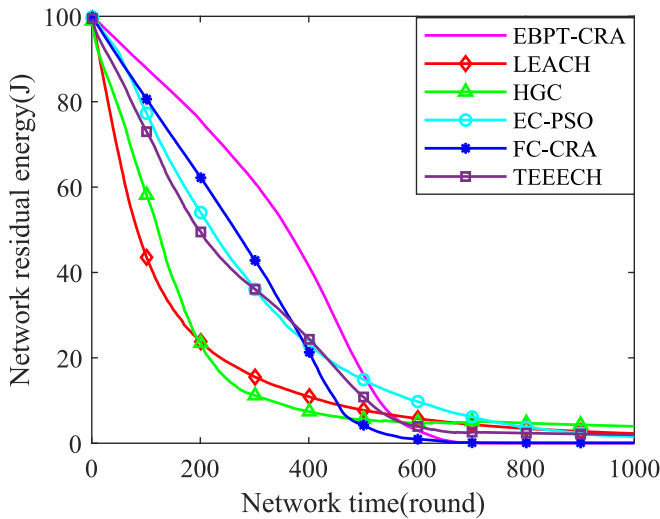


Fig. 8. Network residual energy.

In Fig. 7, The EBPT-CRA fully considers the node load and energy conditions to select the CHs based on the EBPT and CHCCs, avoiding the emergence of extremely large clusters. At the same time, the EBPT-CRA achieves energy balance in inter-cluster communication by adding the

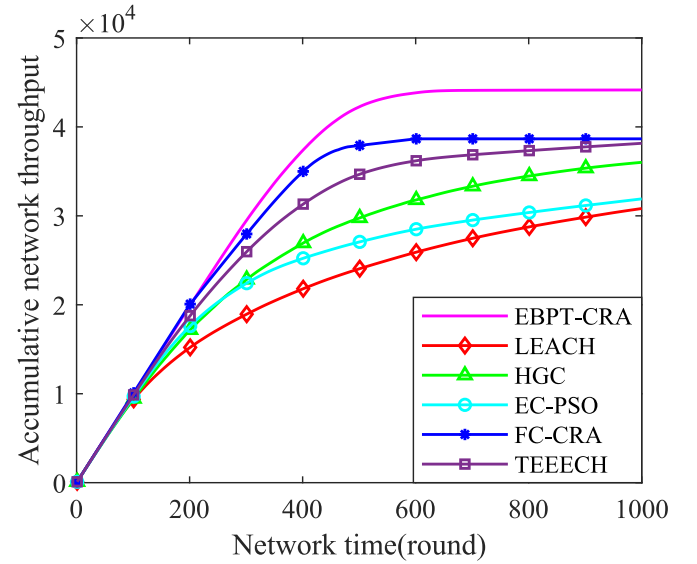


Fig. 9. Accumulative network throughput.

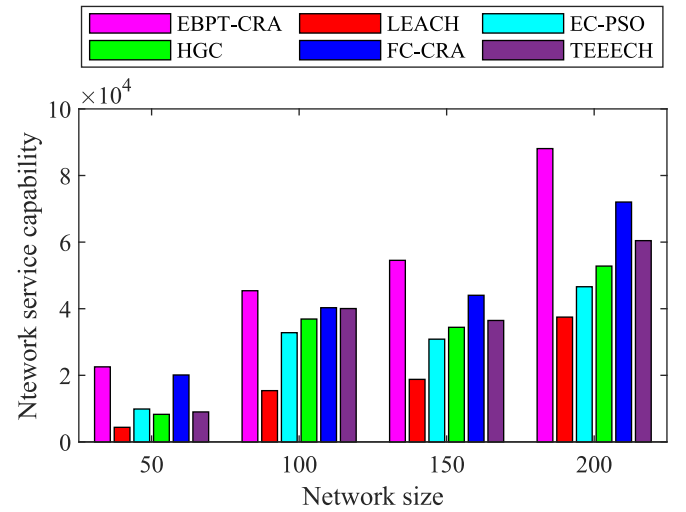


Fig. 10. NSC in different network size.

forwarding nodes. The above measures effectively alleviate the “energy hole” problem, so that the appearance of the first dead node is extended to 200 rounds which is significantly better than the LEACH, HGC and EC-PSO algorithms. The NL under the EBPT-CRA is about 500 rounds, which is the maximum among the five algorithms. The NL and the appearance time of the first dead node are similar between our previous algorithm FC-CAR and the EBPT-CRA, but the number of dead nodes under the FC-CRA is always higher than that under the EBPT-CRA due to the traditional idea of firstly clustering limits the optimization degree of energy consumption.

Due to the random selection of CHs in the LEACH algorithm, the balance of the network energy consumption is not so good, resulting in the first dead node appearing soon after the network runs, and thus the NL is lower than that of other comparison algorithms, which only is about 190 rounds.

The EC-PSO algorithm selects CHs based on the center of mass in the first 50 rounds, and then updates the clusters through the PSO algorithm, which reduces the randomness of CH selection. However, in a large-scale WSN, the distance between CHs is long and the amount of forwarded data at near-BS CHs is large, leading to the aggravation of the “energy hole” problem, and thus the NL is only about 250 rounds.

The HGC algorithm constructs the clusters in similar size based on the given number of CHs, which improves the balance of CH energy, so its curve is better than the LEACH and EC-PSO in Fig. 7. However, the HGC algorithm does not take into account the distance from CH to BS, causing that the CHs close to the BS bearing a large data load form the CHs far from the BS. As a result, the death rate of near-BS CHs remains faster and the performance of the HGC is clear lower than that of EBPT-CRA. The NL of the HGC algorithm is about 360 rounds.

The TEECH algorithm optimizes the probability threshold of the CH selection in LEACH algorithm considering the node energy and node-to-BS distance, which can better balance the node energy, so in Fig. 7, its performance is better than the LEACH, HGC and EC-PSO algorithms. However, its single-hop communication mechanism leads to the premature death of the far-BS CHs due to the huge energy consumption for data transmission, so its performance is lower than the FC-CAR and EBPT-CRA algorithms. The NL of the TEECH algorithm is about 370 rounds.

In Fig. 8, the EBPT-CRA preforms intra-cluster communication based on the paths in the EBPT, which greatly avoids backward data transmission, so the network residual energy in the first 500 rounds is significantly higher than that of the comparison algorithms. From 500 to 600 rounds, the network residual energy decreases faster because the number of surviving nodes is small, the network node density becomes low, and long-distance communication is inevitable.

In the LACH algorithm, both the intra-cluster communication and inter-cluster communication are single-hop communication, which leads to long distance transmission of data in a large-scale WSN, so the network residual energy is smaller than other comparison algorithms in the first 200 rounds. After 200 rounds, the rate of network residual energy reduction decreases because most of the surviving nodes in the network are near-BS nodes and there are not many of them.

In the HGC, the cluster area will become larger due to the decrease of surviving nodes, and the single-hop communication will consume higher energy for data transmission in a cluster. The TEECH algorithm forms clusters by a modified fuzzy C-mean algorithm, which saves the energy for intra-cluster communication. Meanwhile, the near-BS nodes have high chance to become CHs, which reduce the energy consumption of data backward transmission, so the network residual energy of the TEECH algorithm is better than the LEACH and HGC algorithms. The performance of the EC-PSO algorithm is similar to the TEECH algorithm. The EC-PSO algorithm defines the optimal single-hop distance to limit the node communication distance, so there is no long-distance communication between nodes, resulting in that the residual energy of the network is higher than that of the LEACH and HGC algorithms. However, this algorithm does not consider the problem of data backward transmission, so the network residual energy is lower than that of the EBPT-CRA and FC-CRA in the first 400 rounds.

When the clusters are identified, data routing receives constraints from the clustering results and is unable to minimize the routing energy consumption, so the network residual energy of the FC-CRA is lower than that of the EBPT-CRA.

Since the EBPT-CRA improving the number of forwarding nodes in the inter-cluster communication and considers both residual energy and forwarding load of nodes in the inter-cluster communication, the energy consumption of the near-BS nodes is effectively balanced and premature disruption of communication paths in the network is avoided. Therefore, in Fig. 9, the ANT under the EBPT-CRA algorithm is superior than the other algorithms, and approximately proportional to the network time in the first 400 rounds. From 400 to 450 rounds, the growth rate of the curve decreases due to the disruption of communication paths one after another caused by the rapid increase of the dead-node number. The ANT reaches the maximum after 550 rounds because there are fewer surviving nodes in the network.

In large-scale WSNs, the single-hop communication strategy of the LEACH algorithm leads to premature death of the far-BS CHs which causes serious data loss problems, so the ANT of the LEACH algorithm

grows at the slowest rate.

In order to avoid the long-distance transmission of data, the EC-PSO algorithm limits the single-hop transmission distance between nodes, so the death of a few nodes in a large-scale WSN will lead to the disruption of intra-cluster communication paths. Therefore, the ANT under the EC-PSO algorithm grows slowly and only higher than that of the LEACH algorithm.

The HGC algorithm promotes the balance of CH load and energy consumption by the uniform clustering, and thus its ANT profile is better than that of the LEACH and EC-PSO algorithms. However, this algorithm does not fully consider the residual energy of CHs, which leads to the premature death of near-BS CHs and the disruption of communication paths, and thus its ANT is significantly lower than that of the EBPT-CRA and FC-CRA.

In the TEECH algorithm, the super nodes and advanced nodes, which have more initial energy than ordinary nodes, have a high probability of becoming CHs, so the CHs have sufficient energy and time to transmit data, resulting in the ANT being greater than that of the LEACH, HGC and EC-PSO algorithms. However, due to the long-distance single-hop communication between the remote CHs and the BS, the remote CHs consume too much energy and die prematurely, which lead to more and more data of remote nodes not being transmitted to the BS after 300 rounds. Therefore, the performance of the TEECH algorithm is lower than the FC-CRA and EBPT-CRA algorithms.

Since the network residual energy of the FC-CRA is lower than that of the EBPT-CRA, the number of disrupted data paths under the FC-CRA is more than that under the EBPT-CRA, resulting in the ANT of the FC-CRA is not the best one among the algorithms.

Some indicators of the four algorithms are shown in Table 5 when the WSN runs 1000 rounds. Compared with the best performance of the comparison algorithms, the EBPT-CRA improves the ANL by 5.4 %, the NL by 12.3 %, the NSC by 14.3 %, the ANT by 19.6 %, and decreases the average energy consumption of collecting one-bit data (AECDD) by 16.6 %.

As a service performance indicator for large-scale WSNs, the NSC integrates the efficiency of network energy usage and the reliability of data transmission, so this paper selects the NSC to verify the generalizability of the EBPT-CRA algorithm. The simulations are conducted on the networks with 50, 100, 150 and 200 nodes, and the results are the average value of 50 simulations, which is shown in Fig. 10.

In Fig. 10, the performance of the EBPT-CRA is less affected by the network size and outperforms other comparison algorithms at all network size because the size and shape of the EBPT can change dynamically with the number, location, and energy distribution of nodes. In large-scale WSNs, the LEACH algorithm has many far-BS CHs that need to transmit data over long distances, resulting in the premature death of the far-BS CHs and a low NSC. Both the EC-PSO algorithm and HGC algorithm do not consider the balance of CH energy in inter-cluster communication, especially the energy consumption of near-BS CHs, so their performance is lower than that of the EBPT-CRA. The performance of the TEECH algorithm is similar to the HGC and EC-PSO algorithm, because the number of advanced nodes and super nodes increase with the network size, improving energy balance, but the long-distance communication like the LEACH algorithm limits the performance of the TEECH algorithm. As the number of nodes increases, data

Table 5
Performance indicators.

	ANL (rounds)	NL (rounds)	NSC (packets)	ANT (packets)	AECDD (nJ)
LEACH	324.1	195.5	1.48×10^4	3.08×10^4	2597.4
HGC	460.6	360.0	2.54×10^4	3.60×10^4	2222.2
EC-PSO	422.5	250.7	2.05×10^4	3.19×10^4	2507.8
TEECH	463.7	371.6	3.64×10^4	3.73×10^4	2144.7
FC-CRA	472.3	450.2	3.71×10^4	3.77×10^4	2122.0
EBPT-CRA	497.8	505.5	4.24×10^4	4.52×10^4	1769.9

routing consumes more and more energy, and saving energy has an increasing impact on the performance of the NSC, so the NSC of the FC-CRA is gradually lower than that of the EBPT-CRA with the increase in the number of nodes.

To observe the effect of network load on the performance of the algorithms, the network load is varied by changing the packet size based on the parameters in Table 4., and the variation of the NSC for each algorithm is shown in Table 6.

The LEACH algorithm is independent of the packet size and thus its performance varies every little, but the NSC is small due to the inadaptation of large-scale WSNs. As the packet size becomes larger, the node energy consumption increases leading to an increase in the node death rate. In this case, the cluster coverage in the HGC algorithm will increase as the number of surviving nodes decreases, resulting in the higher energy consumption for intra-cluster communication and a significant decrease of the NSC. The CHs in the EC-PSO algorithm is selected dynamically with the change of network energy distribution, but the single-hop distance is fixed, so when the node energy is insufficient, the transmission of many long data packets is interrupted and a large amount of data is lost, resulting in the NSC is significantly affected by the network load and gradually lower than the other algorithms.

The TEECH algorithm stratifies the nodes according to the energy. As the packet size increases and the transmission energy increases, the number of nodes in the low energy layer increases faster and the packet loss is more, resulting in a faster decrease in the NSC than the FC-CRA and the EBPT-CRA. The FC-CRA adjusts the cluster radius and size based on the network energy state, thus the NSC is less affected by the network load and stays at a better level. However, the FC-CRA favors the balance of cluster size, and the intra-cluster energy consumption is low, but the inter-cluster routing energy consumption is not fully considered. Routing energy consumption is the main one in large-scale WSNs, so the EBPT-CRA adjusts both routing and clustering in a timely manner considering routing energy and cluster size, which leads to that the EBPT-CRA is less affected by the network load and the NSC always outperforms the other algorithms.

We observe the effect of various network size (number of nodes) on the performance of the EBPT-CRA with a packet size of 5000bit (a heavy load) in the expectation of finding the optimum number of nodes or maximum number of nodes in the network, and the simulation results are shown in Fig. 11 and Fig. 12.

In Fig. 11, when the network size reaches 150 nodes, the NL reaches the maximum value of 286 rounds, indicating that there is no monotonically increasing relationship between network energy consumption and network size. When the network size is small, although the network load is small, the transmission distance is large, and the network energy is mainly consumed in long-distance transmission. When the network size is large, the network load is heavy, and the network energy is mainly consumed in the processing and forwarding of data. Therefore, the network size should not be too large or too small.

In Fig. 12, the NSC shows a monotonically increasing trend with the network size. Although the NL decreases after the network reaching the maximum value of the NL at the size of 150 nodes, the NSC still increases, but the growth rate gradually decreases. The NSC is jointly determined by NL and network throughput per round. When the network size becomes larger, the increase in network load leads to a

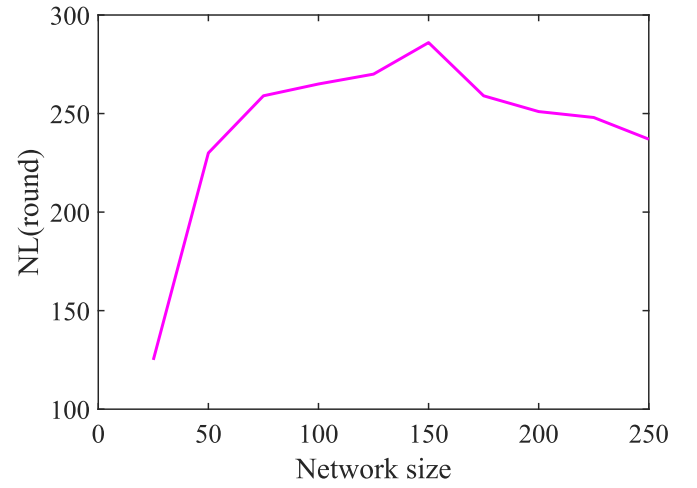


Fig. 11. NL of the EBPT-CRA with heavy load.

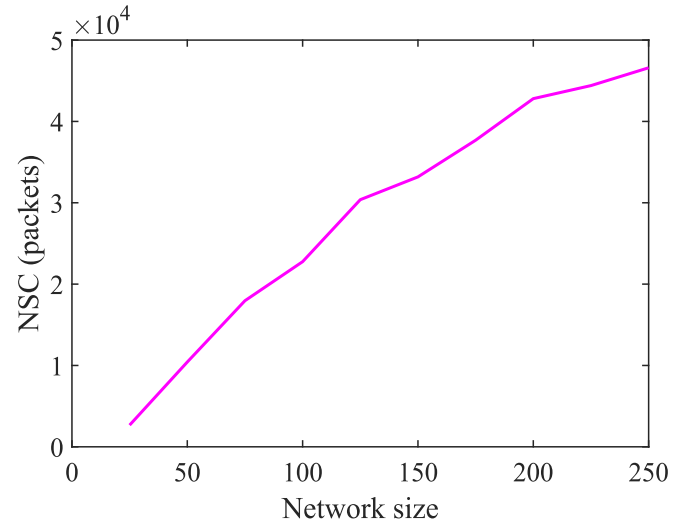


Fig. 12. NSC of the EBPT-CRA with heavy load.

faster node death and a decrease in NL, but the amount of data transmitted per round within NL increase faster, which results in a faster increase in network throughput per round than the decrease in NL (the number of total rounds), leading to the monotonically increasing trend of the NSC.

6. Conclusion

In order to fully utilize the network energy and further alleviate the problem of “energy hole” in large-scale WSNs, this paper introduce the idea of firstly routing and secondly clustering, and proposes a clustering and routing algorithm based on the EBPT, i.e., EBPT-CRA. The algorithm constructs CTs based on the NCB and CHCC, which limits the cluster size and avoids premature death of the CHs due to excessive cluster size, optimizing the network energy distribution. The construction of CTs simultaneously implements CH selection, cluster construction and intra-cluster communication routing, reducing the number of control messages and transmission energy consumption in the network. The intra-cluster nodes perform intra-cluster routing based on the EBPT, which avoids backward data transmission, reduces intra-cluster communication energy consumption, and improves the balance of energy distribution and the reliability of data transmission of the intra-cluster nodes. The near-BS nodes are set as forwarding nodes to participate in the inter-

Table 6
NSC($\times 10^7$ bit) for each algorithm under different packet size.

Packet size (bit)	2500	3000	3500	4000	4500	5000
LEACH	3.7	3.2	3.9	3.4	3.8	3.9
HGC	6.4	6.3	5.8	2.95	2.7	1.9
EC-PSO	5.1	4.0	3.9	2.0	1.9	1.4
TEECH	9.1	7.4	6.4	5.76	5.8	4.8
FC-CRA	9.3	8.6	7.5	6.6	6.3	6.3
EBPT-CRA	10.6	10.7	9.3	9.1	8.5	8.4

cluster routing, which share the data forwarding pressure of the near-BS CHs and alleviate the “energy hole” problem.

The simulation results shows that the EBPT-CRA can effectively save network energy and improve the ANL, NSC and ANT, achieving the optimal utilization of network energy in large-scale WSNs.

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CRedit authorship contribution statement

Bing Fan: Conceptualization, Data curation, Funding acquisition, Methodology, Validation, Writing – review & editing. **Yanan Xin:** Conceptualization, Formal analysis, Investigation, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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